

UNITED STATES PATENT APPLICATION

**SYSTEM AND METHOD FOR AUTOMATED MANUFACTURING
COST ESTIMATION FROM THREE-DIMENSIONAL COMPUTER-
AIDED DESIGN GEOMETRY USING HYBRID MACHINE LEARNING
AND RULE-BASED INFERENCE**

FIELD OF THE INVENTION

Computer-Aided Manufacturing • Cost Estimation • Machine Learning

Application Filing Date: March 21, 2026

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Attorney Docket No.: MWT-2026-001

ABSTRACT OF THE DISCLOSURE

A computer-implemented system and method for automatically estimating manufacturing costs from three-dimensional computer-aided design (CAD) geometry data. The system receives a CAD file in one or more standard interchange formats (STEP, STL, IGES, or native parametric formats), parses the file into a boundary representation (B-Rep) mesh comprising triangulated face elements, and extracts a comprehensive set of geometric features including per-face surface area, unit normal vectors, discrete curvature estimates (mean and Gaussian), surface classification (planar, cylindrical, spherical, conical, toroidal, or freeform), wall thickness measurements, hole detection and characterization, and topological connectivity. The extracted features are encoded as a node-attributed face-adjacency graph suitable for graph neural network (GNN) processing. A hybrid inference engine combines a trained GNN model for manufacturability scoring with a deterministic rule-based cost model that maps geometric features to manufacturing cost drivers including material consumption, machining time, tooling requirements, and risk-adjusted labor. The system produces a real-time cost estimate with itemized breakdown, manufacturability score, risk flags, and actionable optimization recommendations, enabling manufacturers to evaluate part producibility and cost within seconds of CAD file upload rather than the hours or days required by conventional manual quoting processes.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to no prior applications and is filed as an original utility patent application under 35 U.S.C. §111(a).

BACKGROUND OF THE INVENTION

1. Technical Field

The present invention relates generally to computer-aided manufacturing (CAM) and, more particularly, to systems and methods for automatically deriving manufacturing cost estimates from three-dimensional CAD geometry using a combination of computational geometry analysis, machine learning inference, and parametric cost modeling.

2. Description of Related Art

Manufacturing cost estimation is a critical function in discrete manufacturing industries, including aerospace, automotive, medical device, and industrial equipment sectors. Accurate cost estimates directly affect competitive bidding, production planning, design-for-manufacturability (DFM) decisions, and supply chain management. Traditional cost estimation methods rely predominantly on manual analysis by experienced manufacturing engineers who inspect CAD geometry, identify manufacturing features (holes, pockets, bosses, fillets, chamfers, thin walls), select appropriate manufacturing processes, estimate cycle times, and aggregate material, labor, tooling, and overhead costs.

These manual methods suffer from several well-documented deficiencies. First, the process is labor-intensive, typically requiring two to four hours per part for complex geometries, creating a bottleneck in the request-for-quotation (RFQ) pipeline. Second, estimates are highly dependent on the experience and judgment of individual engineers, leading to inter-rater variability of 15–40% for geometrically complex parts. Third, manual processes do not scale: a manufacturer receiving hundreds of RFQs per month cannot apply deep engineering analysis to each, resulting in either superficial estimates or selective quoting that may exclude profitable opportunities.

Prior art systems have attempted to address these limitations through parametric cost estimation software, feature-based costing tools, and more recently, machine learning approaches. Parametric systems (e.g., SEER-MFG, Costimator) require extensive manual

configuration of process parameters and do not directly consume raw CAD geometry. Feature recognition systems (e.g., those integrated into certain CAM platforms) can identify standard manufacturing features but typically cannot handle freeform or non-standard geometries and lack the ability to assess global manufacturability or predict risk. Machine learning approaches, while promising, have generally treated the problem as a regression on aggregate geometric statistics (volume, surface area, bounding box) without leveraging the topological structure of the part geometry, resulting in limited accuracy for complex parts.

There remains a need in the art for a system that (a) directly consumes standard CAD file formats without manual preprocessing, (b) extracts both local and global geometric features including curvature, topology, and manufacturability-relevant characteristics, (c) leverages the inherent graph structure of mesh geometry through graph neural networks, (d) combines learned representations with deterministic engineering rules for reliable cost decomposition, and (e) operates in real time to enable interactive design-cost exploration.

SUMMARY OF THE INVENTION

In accordance with the present invention, a computer-implemented system for automated manufacturing cost estimation comprises: a geometry ingestion module configured to receive a three-dimensional CAD file and parse it into a triangulated boundary representation mesh; a feature extraction engine configured to compute, for each triangular face element of the mesh, a feature vector comprising surface area, unit normal vector, centroid coordinates, discrete mean curvature, discrete Gaussian curvature, minimum and maximum principal curvature estimates, and a surface classification label; a graph construction module configured to build a face-adjacency graph wherein each node corresponds to a face element and each edge corresponds to a shared mesh edge between adjacent faces, the graph being annotated with node features from the feature extraction engine and edge features including dihedral angle, concavity flag, and shared edge length; a hybrid inference engine comprising a graph neural network model trained to predict a manufacturability score from the node-attributed face-adjacency graph and a deterministic rule-based cost model that maps extracted geometric features to manufacturing cost drivers; and an output generation module configured to produce a structured cost estimate comprising

itemized cost breakdown, manufacturability score, risk assessment, and optimization recommendations.

In a preferred embodiment, the feature extraction engine further computes aggregate geometric statistics including total surface area, enclosed volume (via signed tetrahedra summation), bounding box dimensions, connected component count, and surface class distribution. The feature extraction engine further identifies manufacturing-relevant characteristics including minimum wall thickness, hole count and length-to-diameter ratios, draft angle compliance, and surface finish requirements derived from curvature analysis.

In another aspect of the invention, the graph construction module employs an integer-hashed edge lookup using Szudzik's elegant pairing function to achieve $O(F)$ construction time complexity where F is the number of triangular faces, enabling real-time graph construction for meshes comprising hundreds of thousands of faces.

In yet another aspect, the hybrid inference engine comprises a user-adjustable parameter interface that permits a user to override default assumptions including material cost per unit weight, machining rate per unit time, and tolerance specification level, with the cost model recomputing in real time upon parameter adjustment.

In a further aspect, the system generates a portable document format (PDF) manufacturing quote comprising branded header, part identification, manufacturability score with visual indicator, itemized cost breakdown table with volume discount calculations, geometric summary, risk assessment with severity classification, recommended design modifications with estimated impact, return-on-investment summary, and estimated production timeline.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

Reference is now made in detail to the preferred embodiments of the present invention, examples of which are illustrated in the accompanying description. While the invention will be described in conjunction with the preferred embodiments, it will be understood that they are not intended to limit the invention to those embodiments. On the contrary, the invention is intended to cover alternatives, modifications, and equivalents that may be included within the spirit and scope of the invention as defined by the appended claims.

I. System Architecture Overview

The system of the present invention comprises five principal modules operating in a sequential pipeline with feedback capabilities: (1) a Geometry Ingestion Module, (2) a Feature Extraction Engine, (3) a Graph Construction Module, (4) a Hybrid Inference Engine, and (5) an Output Generation Module. Each module is described in detail below.

II. Geometry Ingestion Module

The Geometry Ingestion Module receives a three-dimensional CAD file uploaded by a user through a web-based interface or application programming interface (API). The module supports multiple standard CAD interchange formats including, without limitation, STEP (ISO 10303), STL (stereolithography), IGES (Initial Graphics Exchange Specification), and native parametric formats from major CAD vendors.

Upon receipt, the module performs format detection and dispatches the file to an appropriate parser. For boundary representation (B-Rep) formats such as STEP, the parser extracts the topological and geometric entities directly. For tessellated formats such as STL, the parser reads the triangulated mesh directly. In all cases, the output of the ingestion module is a triangulated mesh represented as an indexed face set comprising: (a) a vertex position array of dimension $[N_v \times 3]$ where N_v is the number of unique vertices, each entry being a three-dimensional Cartesian coordinate; and (b) a face index array of dimension $[N_f \times 3]$ where N_f is the number of triangular faces, each entry being a triplet of vertex indices.

The ingestion module further performs validation checks including: degenerate triangle detection (faces with zero area), non-manifold edge detection, and vertex deduplication with configurable tolerance. For non-indexed geometries, the module generates an explicit index buffer. The module computes vertex normals using area-weighted averaging of adjacent face normals.

III. Feature Extraction Engine

The Feature Extraction Engine operates on the indexed triangulated mesh produced by the Geometry Ingestion Module and computes a comprehensive set of per-face and aggregate geometric features.

A. Per-Face Features. For each triangular face f_i ($i = 0, 1, \dots, N_f - 1$), the engine computes:

(i) Surface area A_i , computed as one-half the magnitude of the cross product of two edge vectors of the triangle.

(ii) Unit face normal vector $\hat{n}_i$, computed as the normalized cross product of two edge vectors.

(iii) Face centroid c_i , computed as the arithmetic mean of the three vertex positions.

(iv) Discrete mean curvature H_i , estimated from the dihedral angles between face f_i and its adjacent faces in the mesh topology. Specifically, for each edge shared between face f_i and an adjacent face f_j , the dihedral angle θ_{ij} is computed as the inverse cosine of the dot product of the respective face normals. The mean curvature is approximated as the average of all such dihedral angles for face f_i .

(v) Discrete Gaussian curvature K_i , estimated from the product of the minimum and maximum dihedral angles among adjacent faces, modulated by a threshold function on the mean curvature to suppress noise in planar regions.

(vi) Principal curvature estimates $\kappa_{\min}$ and $\kappa_{\max}$, derived from the extremal dihedral angles.

(vii) Surface classification label, determined from the principal curvature signature according to the following decision logic: if both $|\kappa_{\min}|$ and $|\kappa_{\max}|$ are below a threshold ϵ (preferably 10^{-4}), the face is classified as planar; if one is below ϵ and the other above, the face is classified as cylindrical; if both are approximately equal (relative difference below 15%), the face is classified as spherical; if both are of the same sign but unequal, the face is classified as conical; if they are of opposite sign, the face is classified as toroidal; otherwise the face is classified as freeform.

B. Aggregate Features. The engine computes the following aggregate statistics over all faces:

- (i) Total surface area, as the sum of all face areas.
- (ii) Enclosed volume, computed using the signed tetrahedra method wherein each triangular face forms a tetrahedron with the origin, and the sum of signed volumes yields the total enclosed volume.
- (iii) Axis-aligned bounding box, defined by the component-wise minimum and maximum vertex coordinates.
- (iv) Surface class distribution, being a histogram of face counts per surface classification label.
- (v) Curvature statistics, including mean Gaussian curvature, mean curvature, maximum absolute curvature, and curvature variance across all faces.
- (vi) Connected component count, determined by breadth-first search on the face-adjacency graph (described infra).

C. Manufacturing Feature Identification. The engine further identifies manufacturing-relevant features:

- (i) Minimum wall thickness, estimated from proximity analysis between opposing face groups.
- (ii) Hole count and characterization, including depth-to-diameter ratio for each detected hole feature.

(iii) Draft angle analysis, identifying faces with insufficient draft angle for mold extraction or casting.

D. Feature Matrix Construction. The per-face features are assembled into a feature matrix X of dimension $[N_f \times 12]$, wherein each row comprises: area, normal vector (3 components), normalized centroid coordinates (3 components, normalized by bounding box diagonal relative to the bounding box center), minimum principal curvature, maximum principal curvature, Gaussian curvature, mean curvature, and relative area (face area divided by mean face area). This matrix serves as the node feature input to the graph neural network.

IV. Graph Construction Module

The Graph Construction Module builds a face-adjacency graph $G = (V, E)$ from the triangulated mesh, where V is the set of face nodes and E is the set of edges representing shared mesh edges between adjacent faces.

A. Topology Traversal Algorithm. The module employs a single-pass, $O(F)$ algorithm for adjacency detection. For each triangular face, the module registers its three mesh edges in a hash map using Szudzik's elegant pairing function for integer edge hashing. Specifically, for a mesh edge defined by vertex indices (a, b) where $a \leq b$, the hash is computed as $h(a,b) = b^2 + b + a$. This produces a unique integer for each unordered vertex pair without the overhead of string concatenation, providing approximately 3 $\times$ speedup over string-key hash maps on meshes exceeding 100,000 faces. When two faces register the same edge hash, an adjacency entry is created.

B. Edge Features. For each adjacency edge between faces f_a and f_b , the module computes:

(i) Dihedral angle, as the inverse cosine of the dot product of the respective face normals, clamped to the range $[-1, 1]$.

(ii) Concavity flag, determined by the sign of the dot product between the cross product of the two face normals and the shared edge direction vector. A negative dot product indicates a concave (interior) junction.

(iii) Shared edge length, as the Euclidean distance between the two shared vertices.

C. Graph Representation. The resulting graph is stored in a compressed format compatible with graph neural network frameworks (specifically, PyTorch Geometric). The edge index is represented as a coordinate (COO) format tensor of dimension $[2 \times 2E]$ (both directions for undirected graph), and the edge attribute matrix is of dimension $[2E \times 3]$, with each row comprising `[dihedral_angle, is_concave, edge_length]`.

D. Subgraph Extraction. The module provides a k-hop subgraph extraction function that, given a seed face index and a hop count k, returns all face indices within k adjacency hops. This is used for localized analysis of regions of interest identified during risk assessment.

V. Hybrid Inference Engine

The Hybrid Inference Engine combines two complementary computational approaches: a learned model for holistic manufacturability assessment and a deterministic model for cost decomposition.

A. Graph Neural Network Model. The GNN component receives the node feature matrix X and the edge index from the Graph Construction Module and produces a scalar manufacturability score in the range $[0, 100]$. The preferred architecture employs multiple message-passing layers (e.g., GraphSAGE or Graph Attention Network layers) followed by a global mean pooling operation and a multi-layer perceptron (MLP) head. The model is trained on a corpus of historically quoted parts with known manufacturability outcomes.

B. Rule-Based Cost Model. The deterministic cost model maps extracted geometric features to four cost categories:

(i) Material cost, computed as a function of enclosed volume, material density, material unit cost (user-adjustable, default \$120/kg for Ti-6Al-4V), and a buy-to-fly ratio estimated from bounding box volume versus part volume.

(ii) Machining cost, computed as a function of total surface area, surface complexity (weighted by curvature statistics and surface class distribution), machining rate (user-adjustable, default \$85/hr), and a tolerance multiplier that scales with the specified tolerance level (standard: 0.85×, precision: 1.0×, ultra-precision: 1.35×).

(iii) Tooling cost, computed as a function of the number of distinct tool types required (estimated from hole count, minimum internal radii, and maximum depth), fixturing complexity (estimated from connected component count and bounding box aspect ratio), and the tolerance multiplier.

(iv) Total cost, as the sum of material, machining, and tooling costs.

C. User-Adjustable Parameters. The inference engine exposes three user-adjustable parameters that permit real-time cost recomputation without re-running the geometric analysis or GNN inference:

(i) Material cost per kilogram, adjustable via a continuous slider control (range: \$40–\$300/kg).

(ii) Machining rate per hour, adjustable via a continuous slider control (range: \$30–\$200/hr).

(iii) Tolerance specification level, selectable from three discrete levels: standard ($\pm 0.1\text{mm}$), precision ($\pm 0.025\text{mm}$), and ultra-precision ($\pm 0.005\text{mm}$).

Upon adjustment of any parameter, the cost model recomputes the affected cost categories and updates the displayed total cost, explanation text, and recommendations in real time (sub-100ms latency for typical part complexity).

D. Explanation Generation. The inference engine generates human-readable explanations for each output dimension. Cost driver explanations are generated by applying threshold-based rules to the extracted geometric features (e.g., “High face count (47) increases machining time and tool path complexity” when face count exceeds 40). Manufacturability issue explanations are generated from feature values that fall outside manufacturing best-practice thresholds (e.g., wall thickness below 1.5mm for titanium alloys). Risk factor explanations are propagated from the risk assessment module.

E. Recommendation Generation. The engine produces prioritized, actionable recommendations by comparing extracted features against manufacturing best-practice thresholds. Each recommendation includes a priority level (high, medium, or low), a specific design action (e.g., “Increase wall thickness from 1.2mm to at least 1.5mm in thin regions”),

and an estimated impact statement (e.g., “Reduces scrap risk by approximately 40%, enables higher feed rates”).

F. Optimization Module. The system includes a multi-objective optimization module that, given an initial geometry graph and material specification, explores a parameter space of geometric modifications to identify configurations that reduce cost while maintaining or improving manufacturability. The optimization employs an evolutionary algorithm (genetic algorithm variant) with configurable population size and generation count, and returns the Pareto-optimal solution with the best cost-manufacturability trade-off. The output includes a direct comparison of original versus optimized cost with percentage savings.

VI. Output Generation Module

The Output Generation Module produces two categories of output: interactive user interface elements and exportable documents.

A. Interactive Output. The system displays results in a side panel comprising: an animated manufacturability score gauge with color coding (green for scores above 80, blue for scores between 60 and 80, amber for scores between 40 and 60, red for scores below 40); an itemized cost breakdown with per-category values; collapsible explanation sections (“Why?” panels) for each output dimension; a real-time parameter adjustment interface; a risk card list with severity-coded badges; a prioritized recommendations list; and a return-on-investment summary showing estimated time savings and cost reduction potential.

B. Document Output. The system generates a PDF manufacturing quote comprising: branded header with company identification, quote number, and date; client information fields; part file identification; manufacturability score with visual progress bar; itemized cost breakdown table with volume discount calculations; geometry summary table; risk assessment table with severity classification; recommended actions table with expected impact; estimated value summary (ROI); estimated production timeline; and professional footer with page numbering and disclaimer.

VII. Error Handling and Graceful Degradation

The system implements comprehensive error handling at each pipeline stage. If CAD parsing fails due to a corrupted or unsupported file, the system displays an informative error

message identifying the failure mode. If feature extraction produces partial results (e.g., due to non-manifold geometry or degenerate faces), the system proceeds with available data and annotates the output with a warning indicator stating “This part is complex—showing partial analysis.” If the GNN model inference fails, the system falls back to a rule-based-only estimate with appropriate disclosure. In all error cases, the system logs structured diagnostic information including file metadata, failure stage, error classification, and stack trace for subsequent debugging and model improvement.

VIII. Progress Indication

The system provides real-time progress indication through a multi-step progress indicator displaying the current pipeline stage. The stages displayed to the user are: (1) Uploading file, (2) Parsing geometry mesh, (3) Extracting features and running GNN, (4) Computing cost estimate, and (5) Generating report. Each stage transitions from a pending state (hollow circle) through an active state (animated spinner) to a completed state (filled checkmark) as the pipeline progresses.

IX. Job History and Persistence

The system maintains a persistent record of all analysis jobs, storing for each job: the original file name, analysis date, computed total cost, manufacturability score, and completion status (completed, partial, or failed). This job history is accessible through the sidebar interface, enabling users to review and compare previous analyses.

CLAIMS

What is claimed is:

1. A computer-implemented method for estimating manufacturing cost from three-dimensional computer-aided design (CAD) geometry, comprising: receiving, by a processor, a CAD file representing a three-dimensional part; parsing the CAD file into a triangulated boundary representation mesh comprising a plurality of triangular face elements and associated vertex positions; computing, for each triangular face element, a feature vector comprising at least surface area, unit normal vector, centroid coordinates, and at least one curvature estimate; constructing a face-adjacency graph wherein each node corresponds to a face element and each edge corresponds to a shared mesh edge between adjacent face elements, the graph being annotated with the computed feature vectors as node attributes and with edge attributes comprising at least dihedral angle between adjacent face normals; processing the annotated face-adjacency graph through a trained graph neural network to produce a manufacturability score; applying a deterministic rule-based cost model that maps the computed geometric features to a plurality of manufacturing cost categories; and generating an output comprising at least the manufacturability score, an itemized cost breakdown, and one or more risk assessments.
2. The method of claim 1, wherein computing a feature vector further comprises computing discrete mean curvature and discrete Gaussian curvature from dihedral angles between the face element and its adjacent face elements.
3. The method of claim 1, wherein computing a feature vector further comprises classifying each face element into a surface class selected from the group consisting of: planar, cylindrical, spherical, conical, toroidal, and freeform, based on principal curvature signature analysis.
4. The method of claim 1, wherein constructing the face-adjacency graph comprises employing an integer-hashed edge lookup using a pairing function to achieve $O(F)$ time complexity, where F is the number of triangular face elements.

5. The method of claim 1, wherein the edge attributes further comprise a concavity flag determined by the sign of the dot product between the cross product of adjacent face normals and the shared edge direction vector.
6. The method of claim 1, further comprising receiving user-adjustable parameters including at least one of: material cost per unit weight, machining rate per unit time, and tolerance specification level; and recomputing the cost breakdown in real time upon adjustment of any user-adjustable parameter without re-executing the graph neural network inference.
7. The method of claim 1, further comprising generating human-readable explanations for each cost category by applying threshold-based rules to the computed geometric features.
8. The method of claim 1, further comprising generating prioritized actionable recommendations by comparing extracted geometric features against manufacturing best-practice thresholds, each recommendation comprising a priority level, a design action, and an estimated impact statement.
9. The method of claim 1, further comprising computing an enclosed volume of the part using a signed tetrahedra summation method, and using the enclosed volume in the deterministic cost model for material cost estimation.
10. The method of claim 1, further comprising: upon failure of any pipeline stage, providing partial results from completed stages together with a notification indicating that full analysis could not be completed.
11. A system for automated manufacturing cost estimation, comprising: a memory storing instructions; and a processor configured to execute the instructions to: receive a three-dimensional CAD file; parse the CAD file into a triangulated mesh; extract per-face geometric features including surface area, normal vectors, curvature estimates, and surface classification labels; construct a face-adjacency graph with node features and edge features; process the graph through a graph neural network to produce a manufacturability score; apply a rule-based cost model to produce an itemized cost breakdown comprising material cost, machining cost, and tooling cost; and output a

structured analysis comprising the manufacturability score, itemized cost breakdown, risk flags, and optimization recommendations.

12. The system of claim 11, wherein the processor is further configured to expose user-adjustable parameters and recompute the cost breakdown upon parameter modification in real time.

13. The system of claim 11, wherein the processor is further configured to generate a portable document format (PDF) quote document comprising branded identification, manufacturability score with visual indicator, itemized cost table, risk assessment, recommended actions, return-on-investment summary, and production timeline estimate.

14. The system of claim 11, wherein the face-adjacency graph is represented in a coordinate (COO) format compatible with graph neural network processing frameworks, with edge indices of dimension $[2 \times 2E]$ for E undirected edges.

15. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform a method comprising: parsing a three-dimensional CAD file into a triangulated boundary representation mesh; extracting, for each face element of the mesh, a multi-dimensional feature vector comprising geometric properties including area, normal orientation, curvature, and surface type classification; building a face-adjacency graph encoding topological relationships between face elements with computed edge features; generating a manufacturability score using a trained graph neural network operating on the face-adjacency graph; computing an itemized manufacturing cost estimate using a deterministic model parameterized by user-adjustable assumptions; generating explanations linking geometric features to cost drivers and manufacturability issues; and presenting the score, cost estimate, explanations, risk assessments, and actionable recommendations through a user interface.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to perform multi-objective optimization to identify geometric parameter modifications that reduce manufacturing cost while maintaining manufacturability, and to display a comparison of original versus optimized cost with percentage savings.

ABSTRACT

A computer-implemented system and method for automatically estimating manufacturing costs from three-dimensional CAD geometry. The system parses CAD files into triangulated meshes, extracts per-face geometric features including curvature, surface classification, and manufacturing characteristics, constructs a face-adjacency graph, and processes it through a hybrid inference engine combining a graph neural network for manufacturability scoring with a deterministic rule-based cost model. User-adjustable parameters enable real-time cost recomputation. The system produces itemized cost breakdowns, risk assessments, explanations, and actionable optimization recommendations, reducing quote turnaround from hours to seconds.